MONESH BOROLE
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EDUCATION

AISSMS IOIT **AUG 2023 – DEC 2027**
Bachelor of Technology in Electronics and Telecommunications | CGPA 8.64

Universal Junior College | HSC | Percentage : 74% **FEB 2023**

St. Thomas Public School | SSC | Percentage : 82% **FEB 2021**

SKILLS

- **Core Electronics**: Analog & Digital Electronics, Signal Conditioning, Power Supply Basics
- **Embedded Systems:** Arduino, ESP8266, PIC18F4550
- **Sensors & Interfacing:** Load Cell (HX711), Ultrasonic Sensor, OLED (I2C)
- **Programming:** C, C++, Embedded C , Python, Linux, Ubuntu
- **Tools & Simulation:** Multisim 11.0, Arduino IDE, PCB Design ,Proteus
- **Hardware Skills:** Circuit Debugging, Hardware Testing, Fault Analysis, Documentation

EXPERIENCE

MINILEC INDIA Pvt. Ltd **JUN 2025 – JUL 2025**
Technical Intern ***Pune,India***

- Executed functional testing and validation of industrial electronic protection devices using defined test procedures and safety checklists.
- Performed fault analysis and documented test observations as per safety standards
- Gained exposure to industrial electronics, compliance, and product lifecycle processes

ASTRONIC TECHNOSYSTEM LLP **DEC 2024 – DEC 2024**
Electronic Intern ***Pune, India***

- Performed hardware debugging using multimeter and basic lab instruments; identified and resolved connection and component-level issues.
- Assisted in circuit validation, testing procedures, and technical documentation for electronics prototypes.
- Gained hands-on experience in electronics testing, including sensor and module-level verification.

PROJECTS

IoT-Based Weighing Scale Machine | ***Embedded C, ESP8266, HX711, Load Cell, Blynk, ThingSpeak*** **OCT 2025**

- Developed a 300 kg IoT-enabled weighing system using a load cell with HX711 amplifier interfaced to ESP8266
- Implemented real-time weight measurement, calibration, and tare/reset functionality for accurate readings.
- Integrated Blynk for live weight monitoring and ThingSpeak for cloud-based data logging and graphical analysis.
- Designed threshold-based alert logic to detect low-weight and abnormal operating conditions.
- Displayed real-time weight values on an OLED display for local visualization.

Smart Zebra Crossing – Automation &Safety System | ***Arduino, Servo Motors, Sensors, Embedded C*** **APR 2024**

- Built an automated pedestrian and traffic safety system using Arduino-based control logic..
- Implemented servo-controlled barricades synchronized with traffic signal states to manage pedestrian crossings.
- Integrated obstacle detection sensors to prevent unsafe barricade operation.
- Designed signal-based automation and safety interlocks to ensure reliable and fail-safe operation.

Transistor-MOSFET IR Object Detection | ***BC547, IRFZ44N MOSFET, IR Sensor, Proteus*** **AUG 2023**

- Designed a microcontroller-free hardware circuit using BC547 NPN transistor for signal inversion and IRFZ44N MOSFET as a low-side switch to drive LED and buzzer on IR object detection.
- Implemented pull-up resistor logic and active-LOW sensor interfacing; validated design via Proteus simulation before zero PCB hardware build.
- Demonstrated low-cost analog hardware design achieving fast response with no processing delay.

Smart Electric Fence Security System | ***ESP32, PIR, GSM (SIM800L), Vibration Sensor*** **MAR 2026**

- Built an IoT-based intrusion detection system using ESP32 with multi-sensor fusion (PIR + vibration + voltage monitoring) to reduce false positives.
- Integrated SIM800L GSM module for real-time SMS alerts and 95%+ detection rate across lab, campus, and agricultural field testing.
- Designed and validated the complete system through Proteus simulation and real-world hardware testing with solar-powered autonomous operation for remote deployment.

Smart Traffic Signal Simulation | ***C, Linux System Programming, IPC, POSIX Threads*** **FEB 2026**

- Developed a real-time smart traffic signal simulation for a four-way junction using C and Linux system programming.
- Implemented inter-process communication using FIFO pipes and message queues for traffic data handling and congestion alerts.
- Designed multithreaded asynchronous logging using POSIX threads, mutexes, and condition variables for synchronized event management.
- Implemented dynamic traffic-based signal control with signal handling and safe shutdown mechanisms.